



Pharmacological outcomes in juvenile myoclonic epilepsy: Support for sodium valproate

Avirup Chowdhury, Martin J. Brodie*

Epilepsy Unit, Western Infirmary, Glasgow G11 6NT, Scotland, UK

ARTICLE INFO

Article history:

Received 16 July 2015

Received in revised form 7 October 2015

Accepted 13 November 2015

Keywords:

Juvenile myoclonic epilepsy

Antiepileptic drugs

Efficacy

Psychiatric comorbidities

Outcomes

Side effects

ABSTRACT

Purpose: Juvenile myoclonic epilepsy (JME) is one of the most frequently diagnosed of the idiopathic generalised epilepsy syndromes, but long term outcome data still remain sparse.

Methods: A retrospective audit was undertaken in 186 patients (male: $n = 78$; female: $n = 108$) diagnosed with JME at the Epilepsy Unit at the Western Infirmary in Glasgow, Scotland between July 1981 and July 2012. Median age at treatment start was 16 years (range 13–44), with median follow-up of 14 years (range 2–32).

Results: Overall, 171 patients (92%) achieved remission with antiepileptic drug (AED) treatment (median 9.5 years; range 1–31). After discontinuing treatment in 28 patients, only 11 remained seizure-free off medication. Fifteen patients (8%) continued to have seizures despite having tried up to 8 AED regimens: (5 male, 10 female), 7 of whom had psychiatric comorbidities. AEDs most commonly prescribed included sodium valproate (VPA; $n = 142$), lamotrigine (LTG; $n = 66$) and levetiracetam (LEV; $n = 22$). More male patients than female attained remission with their first or second AED schedule (88% versus 56%). More male patients (44%) received VPA than female (31%) overall. Fewer male patients than female received LTG (26% versus 74%) and LEV (22% versus 78%). Of the monotherapies, remission was achieved using VPA ($n = 74$; 52%), LTG ($n = 21$; 32%) and LEV ($n = 12$, 55%). A total of 76 (25%) of AED schedules resulted in intolerable side-effects, including 29 with VPA, 12 with LTG and 4 with LEV.

Conclusion: Overall, JME showed a high rate of seizure freedom with AED treatment. VPA appeared to be the most effective AED. Women tended to have a worse outcome than men, since they were increasingly less likely to receive VPA. Patients with psychiatric comorbidities also had a poorer prognosis.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

Juvenile myoclonic epilepsy (JME) is one of the most frequently diagnosed idiopathic generalised (genetic) epilepsy syndromes (IGES) in teenagers with a prevalence of around 18% of IGES and 5–10% of all epilepsies (Jallon and Latour, 2005). It is characterised by myoclonic jerks, tonic-clonic seizures and, on occasion, generalised absences (Camfield et al., 2013). JME has a slight female predominance, with a peak age of onset between 12 and 18 years. Around 30% of patients show photoparoxysmal responses on a surface electroencephalogram (EEG). Young patients with JME may have lifestyle issues that have the potential to increase the likelihood of further seizures, such as sleep deprivation, exposure to stroboscopic flashes, bout drinking of alcohol and recreational drug

use, all with the potential of contributing to suboptimal adherence to antiepileptic drug (AED) therapy (Kasteleijn-Nolst Trenité et al., 2013) and an increased propensity to develop psychiatric co-morbidities (Tellez-Zenteno et al., 2007).

Patients with JME tend to respond well to medication (Striano and Belcastro, 2013). Several hospital-based studies have considered long term outcomes in adolescents with JME, reporting five-year seizure remission rates ranging between 60% (Senf et al., 2013) and 86% (Geithner et al., 2012). In particular, sodium valproate (VPA) is regarded as the most effective AED, but concerns remain regarding teratogenicity (Hitiris and Brodie, 2005; Marson et al., 2007; Crespel et al., 2013). Official guidelines issued by the Medicines and Healthcare Products Regulatory Agency (MHRA) warn against the use of VPA in female patients unless all other appropriate treatments have failed (MHRA, 2015). This study considered whether any differences in gender outcomes arose as a result in a cohort of adolescents and adults with JME diagnosed, treated and followed up at a single centre over a 30-year period.

* Corresponding author. Tel.: +44 141 211 2534; fax: +44 141 211 2072.
E-mail address: Martin.Brodie@glasgow.ac.uk (M.J. Brodie).

2. Materials and methods

Patients included in the analysis had a clinical diagnosis of JME, sometimes supported by an abnormal routine electroencephalogram (EEG), made at the Epilepsy Unit at the Western Infirmary in Glasgow between July 1982 and July 2012 with a minimum of two years of follow-up. The patient was excluded from the analysis if there was any subsequent doubt as to the correctness of the diagnosis. Those who consistently missed clinic appointments and/or were persistently non-adherent with medication were also excluded from analysis. Overall, 186 patients diagnosed with JME at the Epilepsy Unit at the Western Infirmary in Glasgow satisfied the requirements for inclusion in the study. At their first clinic attendance, a structured interview and protocol were used to gather information regarding seizure type, lifestyle factors and family history. At each subsequent visit, changes to the AED regimen, as well as number of seizures and side-effects noted by the patient were recorded in research case-notes, which were stored on-site (Brodie, 2013).

Those patients, who had no further seizures for at least the previous 12 months on an unchanged AED regimen, were classified as being in remission. In some of these, treatment was withdrawn at the patient's request. If seizures recurred in a patient who initially achieved a 12-month or more seizure-free period, they were regarded as having relapsed. Patients were categorised as being uncontrolled if they continued to experience seizure activity of any kind, including myoclonic jerks, despite taking their latest AED treatment schedule appropriately. Other endpoints included years of remission, responses to different drug schedules, and numbers of patients whose medication was changed due to side-effects. Drug choices and treatment responses were compared between male and female patients. A further area of interest was outcomes in patients with treated psychiatric comorbidities at first presentation and over the course of follow-up.

3. Results

Of the 186 patients included this analysis, 42% were male and 58% were female (Table 1). The median age at commencement of treatment was 16 years (range 12–44 years). Median duration of follow-up was 14 years (range 2–32 years). The most common

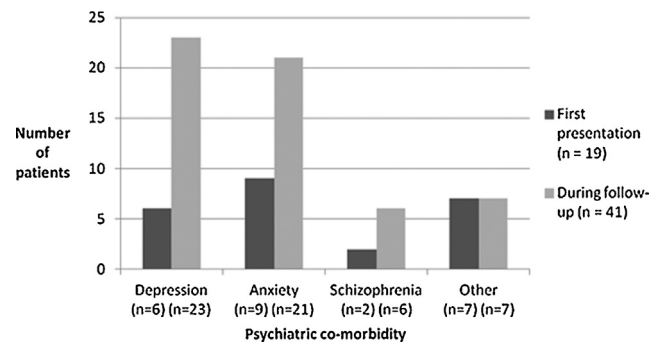


Fig. 1. Psychiatric comorbidities requiring pharmacological intervention at first presentation and during follow-up.

presenting seizure type was generalised tonic–clonic in 72% of patients, with 17% reporting just myoclonic jerks and 11% initially diagnosed with absence seizures before developing myoclonic jerks and/or generalised tonic–clonic seizures. Overall, 32% patients had a first degree relative with epilepsy, while 11% had a history of birth trauma. Febrile convulsions in infancy had occurred in just 6% of patients. Over the course of follow-up, 26% of the population admitted to bout drinking of alcohol while 12% took recreational drugs. A total of 174 patients had a routine EEG, 126 (68%) of whom demonstrated clear epileptiform discharges. Only 64 (34%) patients displayed a photoparoxysmal response. In 12 cases, the patient declined an EEG, and in 27 cases photoparoxysmal response was not tested because of the patient's concern regarding the possible triggering of a seizure. At the time of presentation, 19 patients (10%) had treated psychiatric comorbidities, rising to 41 patients (22%) over the course of follow-up (Fig. 1). The most common diagnoses were anxiety (5% rising to 11%), depression (3% rising to 12%) and schizophrenia (1% rising to 3%). These co-morbidities sometimes coexisted with the original 19 patients having 24 psychiatric diagnoses. The final 41 affected patients had had a total of 57 psychiatric diagnoses made.

In total, 171 patients (92%) achieved remission (Table 2) with a median seizure-free period of 9.5 years (range 1–31 years). Overall 81 patients achieved seizure freedom on their first AED. A further

Table 1
Patient demographic data.

Demographic data	Patients (n = 186)
Gender	
Male	78 [42%]
Female	108 [58%]
Median age at treatment start (years)	16 (range 12–44)
Median duration of follow-up (years)	14 (range 2–32)
First seizure type	
Generalised tonic–clonic	134 [72%]
Myoclonic	31 [17%]
Absence	21 [11%]
Family history of epilepsy	59 [32%]
Birth trauma	20 [11%]
History of childhood absences	17 [9%]
History of febrile convulsions	12 [6%]
Bout drinking of alcohol	49 [26%]
Recreational drug use	22 [12%]
Epileptiform discharges on electroencephalogram	
Present	126 [68%]
Absent	48 [26%]
Not tested	12 [6%]
Photoparoxysmal response on electroencephalogram	
Present	64 [34%]
Absent	95 [51%]
Not tested	27 [15%]

Table 2
Remissions and remission rates (%) for each drug schedule.

Antiepileptic drug/Combination	Number of remissions [%]
Sodium valproate	74 [52%]
Lamotrigine	21 [32%]
Levetiracetam	12 [55%]
Topiramate	6 [67%]
Lamotrigine/Levetiracetam	5 [42%]
Phenobarbital	3 [60%]
Tiagabine	3 [75%]
Zonisamide	3 [50%]
Clonazepam	3 [60%]
Ethosuximide	2 [67%]
Phenytoin	1 [50%]
Valproate/Lamotrigine	14 [52%]
Lamotrigine/Zonisamide	3 [60%]
Topiramate/Lamotrigine	3 [50%]
Valproate/Levetiracetam	3 [23%]
Valproate/Topiramate	3 [38%]
Valproate/Zonisamide	3 [43%]
Valproate/Clobazam	2 [67%]
Levetiracetam/Rufinamide	1 [100%]
Levetiracetam/Zonisamide	1 [33%]
Valproate/Clonazepam	1 [50%]
Valproate/Lamotrigine/Levetiracetam	1 [100%]
Valproate/Zonisamide/Clobazam	1 [100%]
Valproate/Levetiracetam/Topiramate	1 [100%]
Zonisamide/Lamotrigine/Levetiracetam	1 [100%]

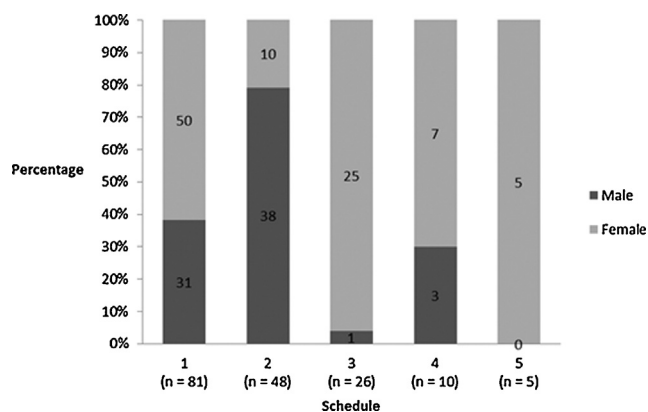


Fig. 2. Gender of patients achieving remission by drug schedule. Numbers inside each section represent the number of patients, while the axis shows the percentage of male and female patients attaining remission with each schedule.

90 patients remained seizure free for at least 12 months on a given AED schedule, but relapsed subsequently regaining seizure freedom on a different regimen. The remaining 15 patients (8%) never became seizure-free for any 12 month period (5 male, 10 female). Overall, 5 of these patients were receiving treatment with their third AED regimen, 2 with their fourth, 4 with their fifth, 1 with their sixth and 3 with their eighth. Treatment was discontinued in a total of 28 patients, 11 of whom remained in remission, whereas the other 17 subsequently relapsed requiring reintroduction of AED therapy.

Of the 105 patients who failed to achieve remission on their first AED, a change of treatment was required due to lack of efficacy in 77%, while the remaining 23% reported intolerable side-effects. Remission at last clinic visit had been achieved by 133 patients on monotherapy, 34 patients on duotherapy, and 4 on three AEDs. In total, 81 patients achieved remission on their first AED schedule, 48 on the second, 26 on the third, 10 on the fourth, 5 on the fifth and 1 on the seventh.

Overall, a higher proportion of the patients who responded to their first or second AED schedule were male than in subsequent schedules (Fig. 2). Similar proportions of males (31/78; 40%) and females (50/108; 46%) responded to their first AED regimen, but more males (38/47; 81%) responded to their second AED than did female patients (10/58; 17%). As a result, 48 female patients, compared to 9 male, remained uncontrolled following their second drug schedule. Both genders tended to relapse and remit equally.

A wide range of AEDs and combinations were used to treat these patients over the 30 years of the study with varying outcome. The most commonly prescribed drugs were sodium valproate (VPA; 142 patients, 52% remission rate), lamotrigine (LTG; 66 patients, 32%), levetiracetam (LEV; 22 patients, 55%), VPA/LTG (27 patients, 52%), VPA/LEV (13 patients, 23%) and LTG/LEV (12 patients, 42%). These 6 AED schedules comprised 71% of all prescribed regimens. Other AEDs were used less frequently. These samples were too small to be able to discern any useful patterns of response.

Similar numbers of male and female patients were treated with a first and second AED schedule. However, most male patients achieved remission early (Fig. 2): 88% of the male population (69 patients), compared to 56% of females (60 patients) achieved remission with their first or second AED schedule. More males than female received VPA as their first AED (53% versus 47%). Overall, similar proportions of male and female patients were treated with VPA (46% versus 54%). Male patients were less likely than female to receive LTG and LEV resulting in gender differences with both as

first AED (LTG: 44% versus 56%; LEV: 22% versus 78%) and across AED schedules (LTG: 26% versus 74%; LEV: 18% versus 82%). VPA prescribing represented 44% of all AEDs given to male patients, compared to only 31% for female patients. Similar proportions of male and female patients were treated with AEDs other than VPA, LTG and LEV (42% versus 43%).

Over the course of follow-up, 76 AED monotherapy schedules (19%) produced intolerable side-effects necessitating a change of medication. VPA monotherapy was discontinued in 29 patients. The main problems were concerns regarding teratogenicity ($n=8$), weight gain ($n=7$) and sedation ($n=5$). The most frequent problems with LTG were rash ($n=3$) and nausea ($n=2$) out of a total of 12 cases. Aggression ($n=4$) was the most common reason for LEV withdrawal. Other AED monotherapies producing intolerable side-effects included TPM with paraesthesia ($n=2$), mood changes ($n=1$), memory/concentration problems ($n=1$) and word-finding difficulties ($n=1$) and ZNS with one patient complaining of intolerable nausea.

Patients with psychiatric co-morbidities tended to fare less well. Seven out of the 15 patients who never achieved remission had treated psychiatric illnesses. Within this group, 5 patients were female, and 2 were male. These patients were tried on a median 5 (range 4–8) AED schedules: 1 was on their fourth, 3 were on their fifth, 1 was on their sixth and 2 were on their eighth at the time of analysis.

4. Discussion

This study explored pharmacological outcomes in a cohort of patients with JME diagnosed and followed-up at a single centre over a 30-year period. Most patients presented in the second decade of life. This analysis demonstrated an overall remission rate of 92% in keeping with those reported in other similar studies (Camfield et al., 2013; Crespel et al., 2013; Baykan et al., 2013). Some patients remained seizure-free despite discontinuing AED therapy. These findings highlight the relatively benign nature of JME in some patients. A few patients with JME diagnosed many years ago were treated initially with inappropriate AEDs, particularly CBZ. These tended to be prescribed between 1981 and 2001 with steadily decreasing frequency. This reflects an improvement in the early diagnosis of JME even in atypical clinical presentations. Although these patients initially gained little benefit from these drugs, most subsequently went into remission on an appropriate AED schedule.

Our data demonstrated that patients with JME often developed psychiatric co-morbidities, which sometimes preceded the diagnosis. These patients tended to require more AED schedules to achieve remission, and were over-represented in the cohort who failed to achieve optimal seizure control. Others have also reported a higher incidence of psychiatric diagnosis among people with epilepsy than in the general population (Lin et al., 2012), as well as poorer outcomes in patients with depression and other psychiatric comorbidities in medical and surgical settings (Hitiris et al., 2007; Kanner et al., 2009; Petrovski et al., 2010). Although psychiatric illness may be a risk factor for pharmacological resistance, it may also be that these patients have more severe subsyndromes of JME (Martínez-Juárez et al., 2006) demonstrating a greater brain dysfunction (Lin et al., 2012). Furthermore, McAuley and co-workers (McAuley et al., 2015) have reported that patients with low mood are less likely to comply with treatment or appropriate lifestyle modifications. This might have been the case for some teenagers with JME in this study, particularly early in the course of their epilepsy.

In this analysis, a higher number of female patients remained uncontrolled than males, and so more female than male patients

required a third or later AED schedule to achieve optimal seizure control. This appeared to be due to a shift away from VPA prescribing in women of childbearing age: 83% of all male patients were prescribed VPA as a first or second AED, compared to only 60% of female patients. Overall, VPA represented 44% of all AEDs prescribed to male patients, compared to 31% for female patients. The latter group tended increasingly to receive LEV or LTG initially in place of VPA due to issues relating to teratogenicity particularly over the past decade (Ackers et al., 2009). In addition, clinical guidelines are reinforcing this approach, with both the Medicines and Healthcare Products Regulatory Agency [MHRA; 11] and the International League Against Epilepsy (ILAE) and European Academy of Neurology (EAN) joint taskforce (Tomson et al., 2015) discouraging the use of VPA in women and girls unless other treatments fail. The ILAE-EAN taskforce also advocate considering a switch away from VPA if a female patient established on treatment wished to consider pregnancy (Tomson et al., 2015).

A proportion of female patients discontinued VPA due to concerns over teratogenicity, the single most common problem reported as “intolerable” by patients. The overall incidence of side-effects leading to drug withdrawal was high, at 19% of all AED prescriptions, with a similar rate of side-effects across AEDs. However, VPA had the widest range of side-effects, including weight gain, hair loss, gynaecomastia and hirsutism; relatively fewer different side-effects were reported for LTG and LEV. This may have been due to higher prescriptions of VPA, but may equally point to an underlying difficulty for patients in tolerating the drug.

Previous studies have highlighted that some patients may not even be offered the choice of taking VPA, resulting in more seizure-related complications (Mole et al., 2015), despite this being a positive predictor of optimal seizure control (Asadi-Pooya et al., 2014). Despite the lack of major double-blind, randomised-control trials for this indication, VPA is widely regarded as the most effective AED for treating JME (Berkovic et al., 2007; Noachter et al., 2008; Machado et al., 2013). This also appeared to be the case in this study. Some small randomised studies have suggested promising alternatives to VPA for JME in LEV (Berkovic et al., 2007; Noachter et al., 2008), LTG (Machado et al., 2013) and TPM (Mantoan and Walker, 2011). Although results with LEV in the relatively small numbers of our patients receiving this AED were promising, this study did not confirm particularly good efficacy for LTG in this population. Sample sizes were too small to assess the value of TPM. However, there may be other reasons for poorer outcomes in female patients. Sex hormones, such as oestrogens, have been shown to be epileptogenic, lowering seizure threshold (Scharfman and MacLusky, 2006). Furthermore, young women may display intermittent adherence with their treatment prior to and during pregnancy because of fear of teratogenicity irrespective of the drug schedule, since no AED can yet be regarded as unequivocally safe for the developing foetus (Hernández-Díaz et al., 2012).

5. Conclusions

This single-centre study of 186 patients with JME took place over a 30 year period in a changing clinical and pharmacological environment. Overall, JME tended to respond well to AED therapy with a 92% remission rate. A sustained rate of remission continued into later AED schedules, although patients with psychiatric co-morbidities tended to do less well. This study supports the suggestion that VPA may be the most effective drug for treating this syndrome, but problems relating to teratogenicity have resulted in a shift away from its prescription in female patients, which could explain their poorer pharmacological outcomes compared to those

in males in this analysis. Similar rates of intolerable side-effects were reported across AEDs, but VPA tended to have the widest side-effect profile. Patient concerns over teratogenicity remained the most commonly reported reasons for its withdrawal.

6. Conflict of interest statement

Avirup Chowdhury has no conflicts of interest to declare. Martin Brodie is a member of the advisory boards for Eisai Inc, UCB Pharma, GlaxoSmithKline, Lundbeck, Bial, GW Pharmaceuticals and Takeda. He is on the speakers' bureau for Eisai Inc, UCB Pharma, GlaxoSmithKline and Lundbeck, and has received support for travel to congresses from UCB Pharma, GlaxoSmithKline and Lundbeck.

References

- Jallon, P., Latour, P., 2005. Epidemiology of idiopathic generalized epilepsies. *Epilepsia* 46 (Suppl. 9), 10–14.
- Camfield, C.S., Striano, P., Camfield, P.R., 2013. Epidemiology of juvenile myoclonic epilepsy. *Epilepsy Behav.* 28 (Suppl. 1), S15–S17.
- Kasteleijn-Nolst Trenité, D.G., Schmitz, B., Janz, D., Delgado-Escueta, A.V., Thomas, P., Hirsch, E., et al., 2013. Consensus on diagnosis and management of JME: from founder's observations to current trends. *Epilepsy Behav.* 28 (Suppl. 1), S87–S90.
- Tellez-Zenteno, J.F., Patten, S.B., Jetté, N., Williams, J., Wiebe, S., 2007. Psychiatric comorbidity in epilepsy: a population-based analysis. *Epilepsia* 48, 2336–2344.
- Striano, S., Belcastro, V., 2013. Treating myoclonic epilepsy in children: state-of-the-art. *Expert Opin. Pharmacother.* 14, 1255–1261.
- Senf, P., Schmitz, B., Holtkamp, M., Janz, D., 2013. Prognosis of juvenile myoclonic epilepsy 45 years after onset: seizure outcome and predictors. *Neurology* 81, 2128–2133.
- Geithner, J., Schneider, F., Wang, Z., Berneiser, J., Herzer, R., Kessler, C., et al., 2012. Predictors for long-term seizure outcome in juvenile myoclonic epilepsy: 25–63 years of follow-up. *Epilepsia* 53, 1379–1386.
- Crespel, A., Gelisse, P., Reed, R.C., Ferlazzo, E., Jerney, J., Schmitz, B., et al., 2013. Management of juvenile myoclonic epilepsy. *Epilepsy Behav.* 28 (Suppl. 1), S81–S86.
- Hitiris, N., Brodie, M.J., 2005. Evidence-based treatment of idiopathic generalized epilepsies with older antiepileptic drugs. *Epilepsia* 46 (Suppl. 9), 149–153.
- Marson, A.G., Al-Kharusi, A.M., Alwaidh, M., Appleton, R., Baker, G.A., Chadwick, D.W., et al., 2007. The SANAD study of effectiveness of valproate, lamotrigine, or topiramate for generalised and unclassifiable epilepsy: an unblinded randomised controlled trial. *Lancet* 369, 1016–1026.
- Medicines and Healthcare Products Regulatory Agency, 2015. Medicines related to valproate: risk of abnormal pregnancy outcomes. *Drug Saf. Update* 8 (Jan (6)), 1.
- Brodie, M.J., 2013. Road to refractory epilepsy: the Glasgow story. *Epilepsia* 54 (Suppl. 2), 5–8.
- Baykan, B., Martínez-Juárez, I.E., Altindag, E.A., Camfield, C.S., Camfield, P.R., 2013. Lifetime prognosis of juvenile myoclonic epilepsy. *Epilepsy Behav.* 28 (Suppl. 1), S18–S24.
- Lin, J.J., Mula, M., Hermann, B.P., 2012. Uncovering the neurobehavioural comorbidities of epilepsy over the lifespan. *Lancet* 380, 1180–1192.
- Hitiris, N., Mohanraj, R., Norrie, J., Sills, G.J., Brodie, M.J., 2007. Predictors of pharmacoresistant epilepsy. *Epilepsy Res.* 75, 192–196.
- Kanner, A.M., Byrne, R., Chicharro, A., Wu, J., Frey, M., 2009. A lifetime psychiatric history predicts a worse seizure outcome following temporal lobectomy. *Neurology* 72, 793–799.
- Petrovski, S., Szoek, C.E., Jones, N.C., Salzberg, M.R., Sheffield, L.J., Huggins, R.M., et al., 2010. Neuropsychiatric symptomatology predicts seizure recurrence in newly treated patients. *Neurology* 75, 1015–1021.
- Martínez-Juárez, I.E., Alonso, M.E., Medina, M.T., Durón, R.M., Bailey, J.N., López-Ruiz, M., et al., 2006. Juvenile myoclonic epilepsy subtypes: family studies and long-term follow-up. *Brain* 129, 1269–1280.
- McAuley, J.W., Passen, N., Prusa, C., Dixon, J., Cotterman-Hart, S., Shneker, B.F., 2015. An evaluation of the impact of memory and mood on antiepileptic drug adherence. *Epilepsy Behav.* 43, 61–65.
- Ackers, R., Besag, F.M., Wade, A., Murray, M.L., Wong, I.C., 2009. Changing trends in antiepileptic drug prescribing in girls of child-bearing potential. *Arch. Dis. Child.* 94, 443–447.
- Tomson, T., Marson, A., Boon, P., Canevini, M.P., Covanis, A., Gaily, E., Kälviäinen, R., Trinka, E., 2015. Valproate in the treatment of epilepsy in women and girls: recommendations from a joint Task Force of ILAE-Commission on European Affairs and European Academy of Neurology (EAN). *Epilepsia* 56 (Apr), 1006–1019.
- Mole, T.B., Appleton, R., Marson, A., 2015. Withholding the choice of sodium valproate to young women with generalised epilepsy: are we causing more harm than good? *Seizure* 24, 127–130.
- Asadi-Pooya, A.A., Hashemzadeh, Z., Emama, M., 2014. Predictors of seizure control in patients with juvenile myoclonic epilepsy (JME). *Seizure* 23, 889–891.

- Berkovic, S.F., Nowlton, R.C., Leroy, R.F., Schiemann, J., Falter, U., On behalf of the levetiracetam No 1057 study group, 2007. Placebo-controlled study of levetiracetam in idiopathic generalised epilepsy. *Neurology* 69, 1751–1760.
- Noachter, S., Andermann, E., Meyvisck, P., Andermann, F., Gough, W.B., Schiemann-Delgado, J., et al., 2008. Levetiracetam for the treatment of idiopathic generalised epilepsy with myoclonic seizures. *Neurology* 70, 607–616.
- Machado, R.A., García, V.F., Astencio, A.G., Cuartas, V.B., 2013. Efficacy and tolerability of lamotrigine in juvenile myoclonic epilepsy in adults: a prospective, unblinded randomized controlled trial. *Seizure* 22, 846–855.
- Mantoan, L., Walker, M., 2011. Treatment options in juvenile myoclonic epilepsy. *Curr. Treat. Options Neurol.* 13, 355–370.
- Scharfman, H.E., MacLusky, N.J., 2006. The influence of gonadal hormones on neuronal excitability, seizures, and epilepsy in the female. *Epilepsia* 47, 1423–1440.
- Hernández-Díaz, S., Smith, C.R., Shen, A., Mittendorf, R., Hauser, W.A., Yerby, M., et al., 2012. Comparative safety of antiepileptic drugs during pregnancy. *Neurology* 78, 1692–1699.